

Implement Gradient Descent Algorithm to minimize given objective function.

Steps / Algorithm for the Experiment

Step 1: Define the Objective Function

- Write a convex function that returns $f(x_1, x_2)$

Step 2: Compute the Gradient (Partial Derivatives)

- Find and define the gradient of the function:

$$\partial f / \partial x_1 \text{ and } \partial f / \partial x_2$$

Write a Python function that returns this gradient as a NumPy array.

Step 3: Implement Gradient Descent Algorithm

- Initialize:
 - Learning rate α
 - Number of iterations
 - Initial values $x=[x_1, x_2]$
- Loop for a fixed number of iterations:
 - Compute gradient at current point
 - Update values:

$$x = x - \alpha \cdot \nabla f(x)$$

$\nabla f(x)$ represents a **vector of all partial derivatives** — one for each variable.

Store each step (for plotting later)

Step 4: Print Final Results

- Print the final values of:
 - x_1, x_2
 - Function value $f(x_1, x_2)$

Step 5: Visualize the Optimization Path

- Create a grid over a range of x_1 and x_2 values
- Compute function values for the grid to generate a **contour plot**
- Overlay the path taken by Gradient Descent (in red)